WINDOWS SERVER 2022+

ACTIVE DIRECTORY

Windows Server Domain & Client Integration

Personal IT Project

Project Focus

Build, configure, administer, and verify a Windows domain environment.

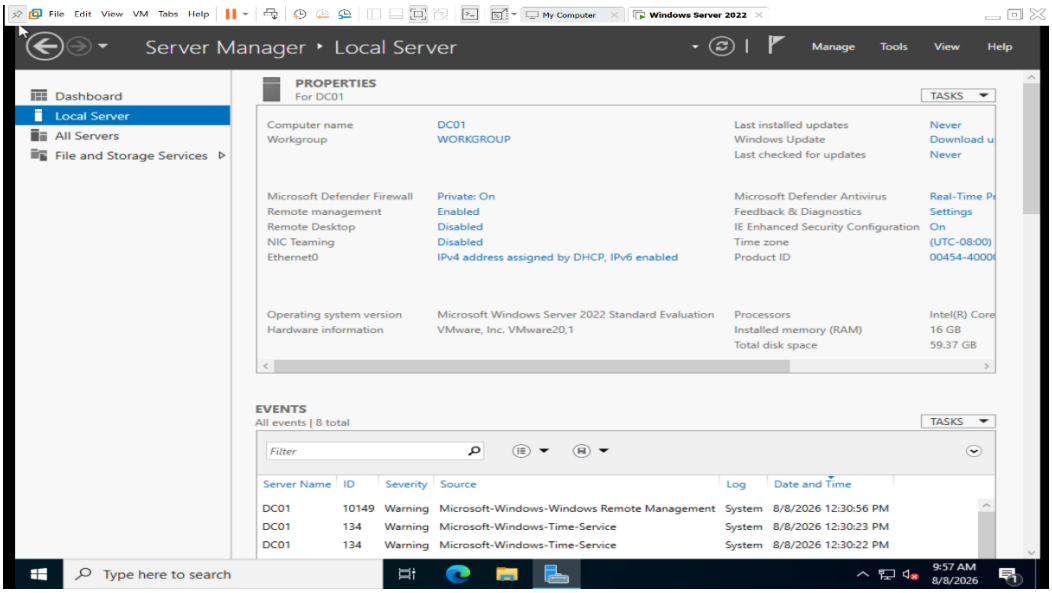
Environment

Windows Server 2022• Windows 11• VMware• Active Directory Domain Services• DNS

Domain	Domain Controller
homelab.local	DC01

Technical Skills

User administration• Password management• Security groups• Organizational Units• DNS verification• Windows troubleshooting



Windows Server 2022 / DC01 environment in Server Manager.

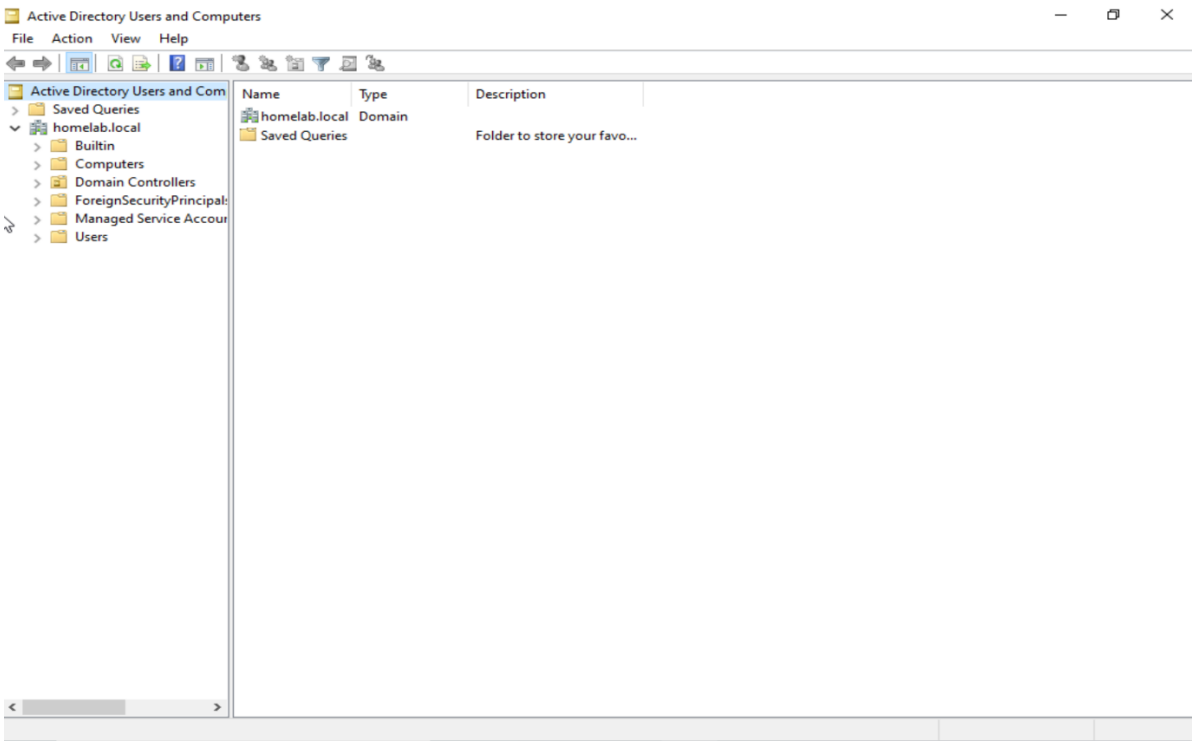
1.Project Overview

I built this lab to learn how Windows domain works and to get more practice with Active Directory. I set up Windows Server 2022 as my domain controller and created the homelab.local domain.

In the lab, I created users and organized them into different department OUs. I also created security groups, added users to those groups, and practiced changing the user's password.

I also set up DNS and used a Windows 11 computer to check that homelab.local could be found on the network.

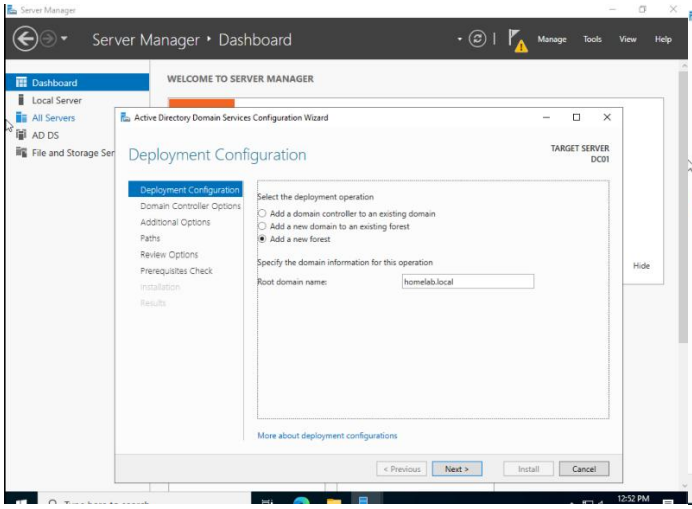
Component	Configuration
Virtualization	VMware
Domain Controller	Windows Server 2022-DC01
Directory	Active Directory Domain Services
DNS	Windows Server DNS
Active Directory Domain	homelab.local
Organizational Units	IT, HR, Finance, Sales, Workstations
Client/ Verification	Windows 11 workstation and DNS resolution



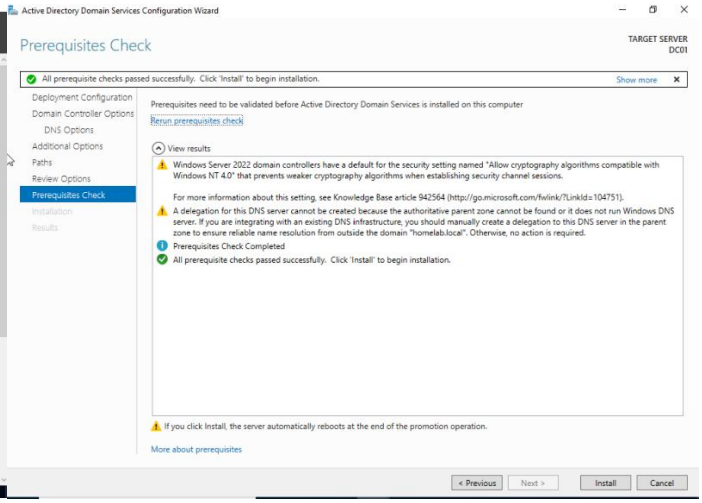
Active Directory Users and Computers showing the homelab.local domain structure

2. Domain Controller Deployment

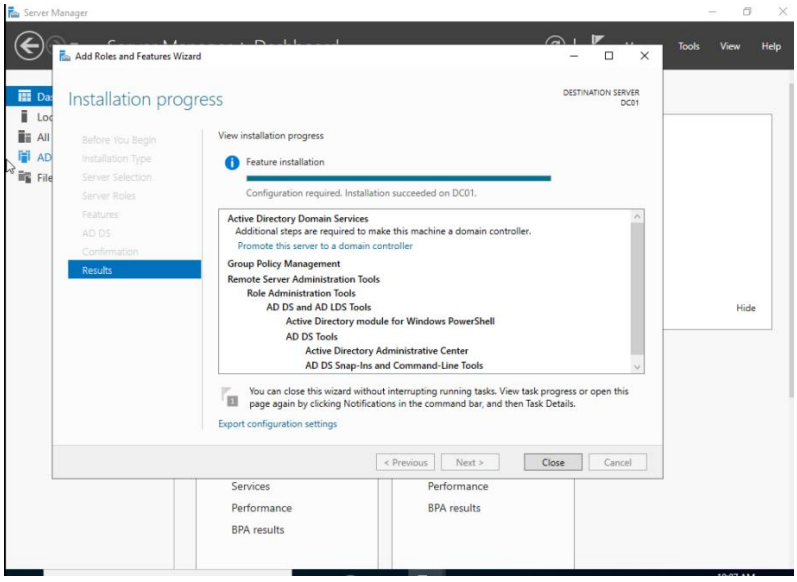
I set up Windows Server 2022 as the domain controller for my lab. I installed Active Directory Domain Services, created the homelab.local domain, and set up DNS during the process.



AD DS Configuration Wizard creating a new forest with homelab.local.



Prerequisites check completed successfully before promotion.

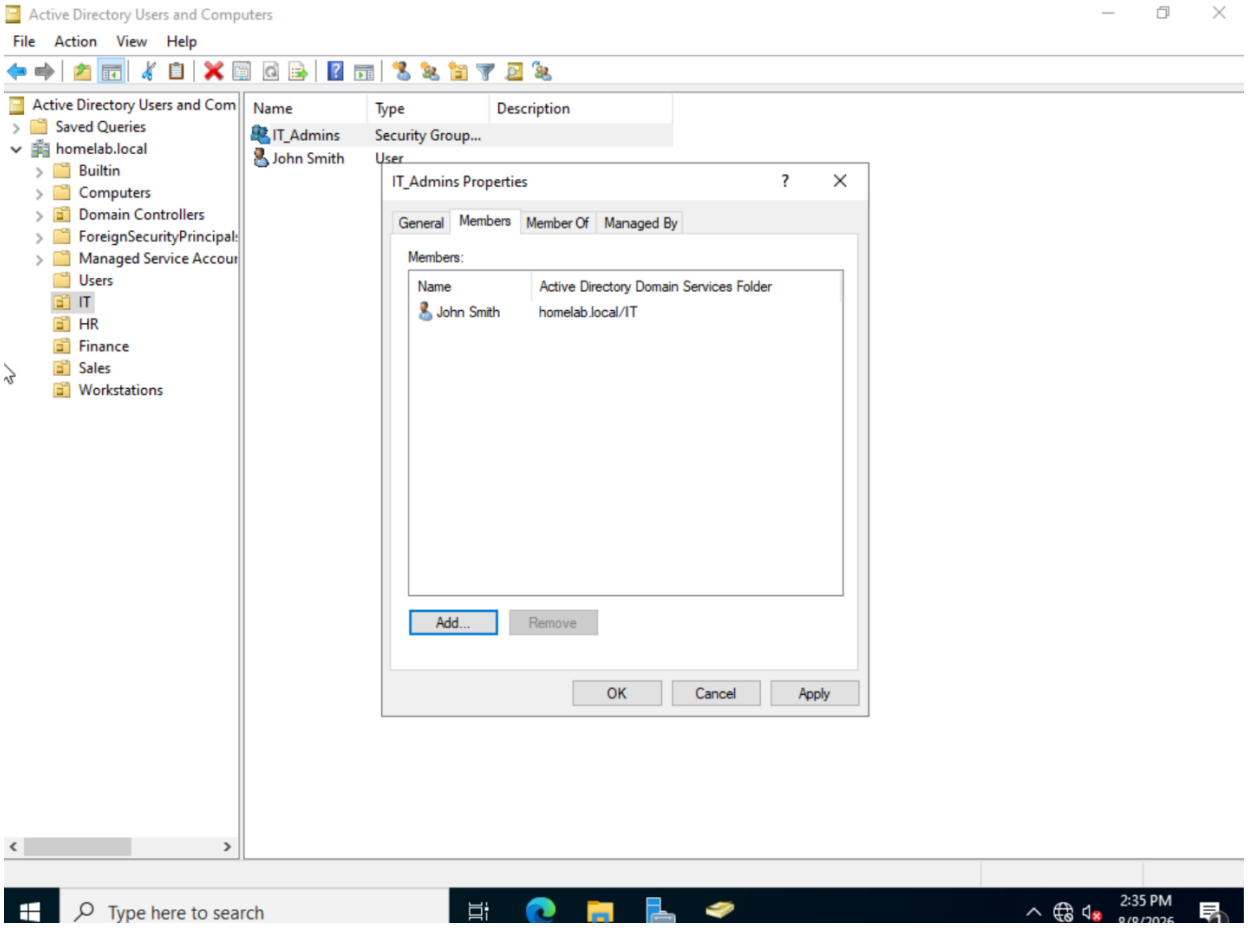


AD DS role installation completed successfully.

3. Active Directory Structure

I organized the domain into separate OUs for each department. This keeps users and computers organized and makes them easier to manage.

Organizational Units	Purpose
IT	IT users and administrative access
HR	Human Resources users
Finance	Finance users
Sales	Sales users
Workstations	Computer objects

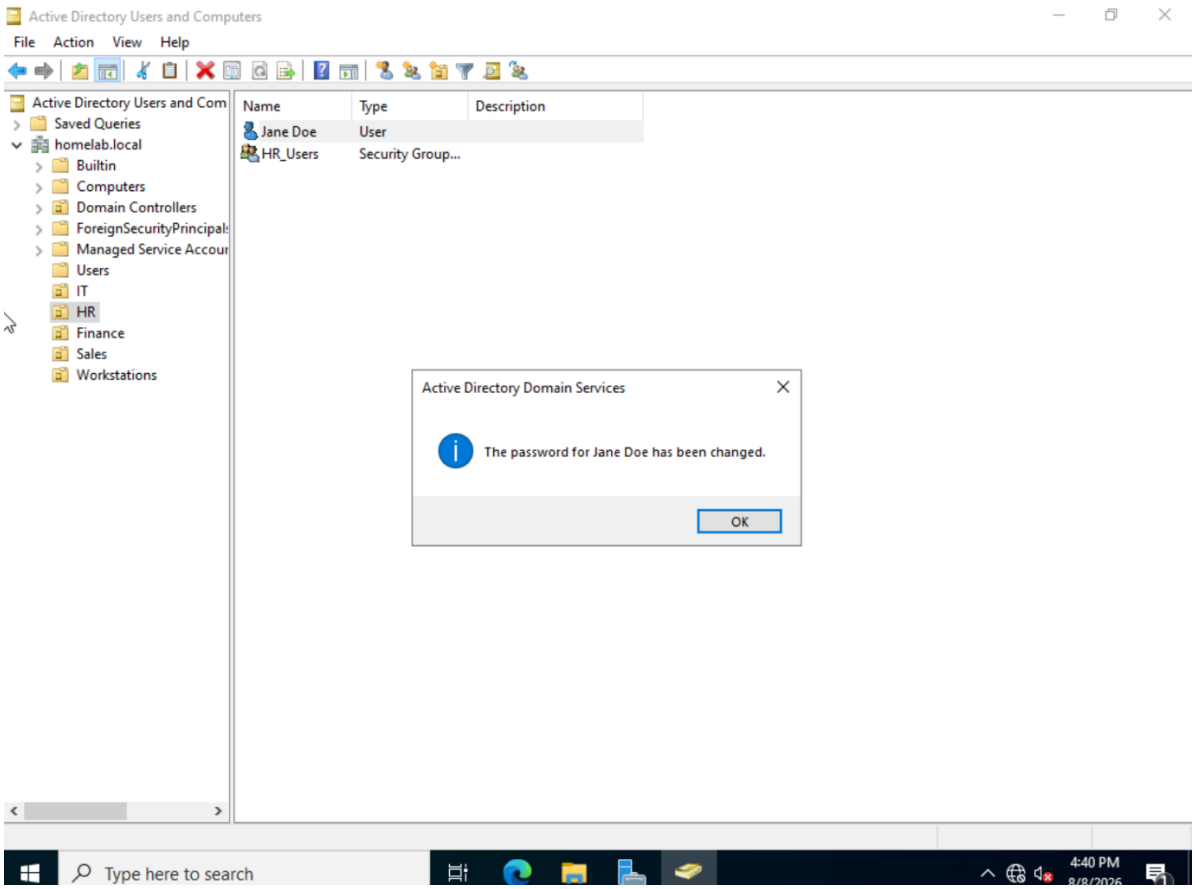


HR OU showing John Smith and the IT_Admins security group.

4.User Account Administration

I used the lab to practice some of the basic tasks that come up when managing user accounts, including creating users, changing passwords, managing group membership, and checking that everything was set up correctly.

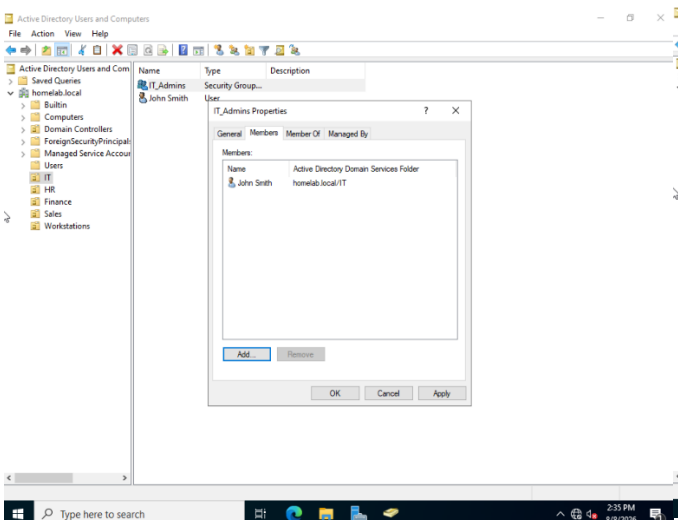
Workflow	What the lab demonstrates	Ticket Connection
Create User	Create a domain account in the correct OU	New hire / onboarding
Reset Password	Change a user's domain password	Forgotten password / sign-in issue
Group membership	Add users to the correct security group	Access / permissions request
Verify	Check the account and DNS setup	Troubleshooting / validation



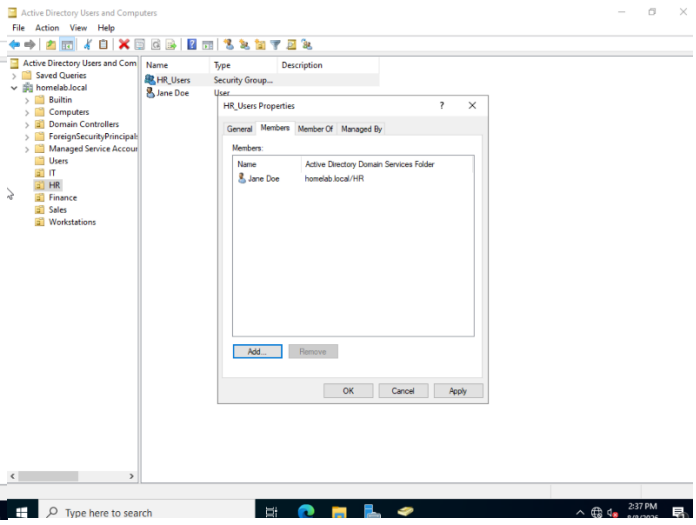
Jane Doe's password was successfully changed in Active Directory.

5. Group Based Access Management

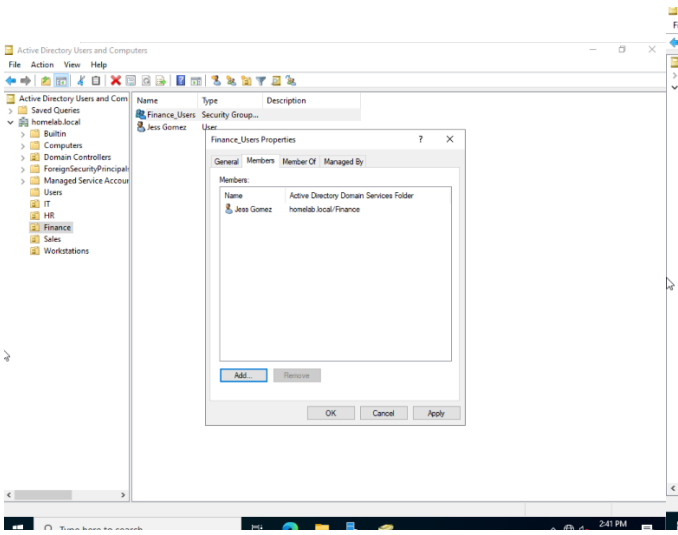
I created security groups for each department and added the users to the group that matched their department. This gives me a simple way to manage access based on the user's role or department.



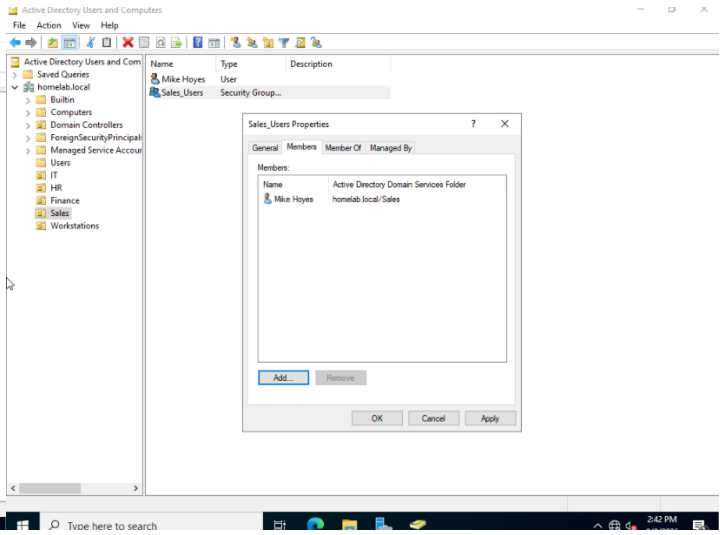
IT_Admins – John Smith



HR_Users- Jane Doe



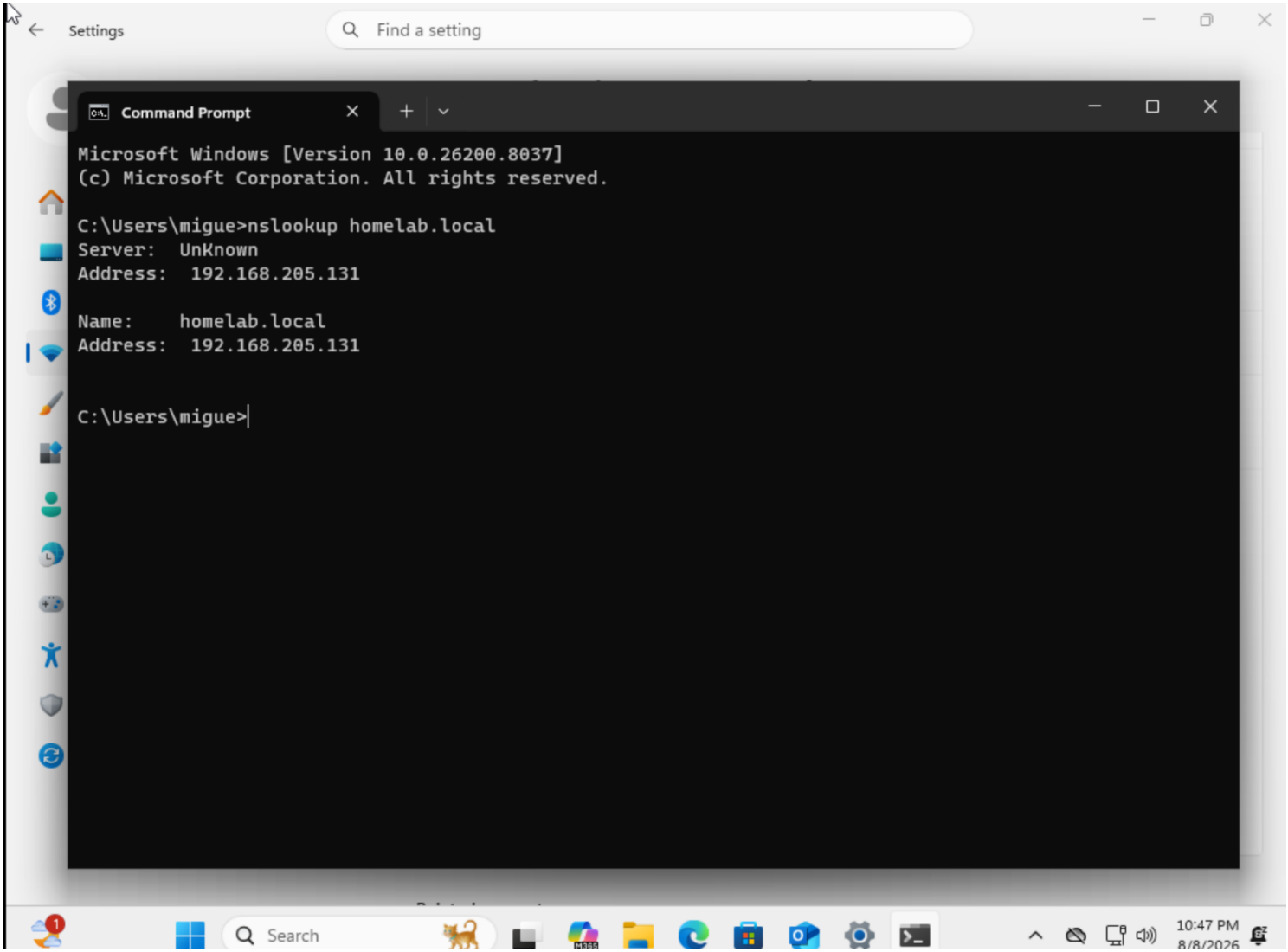
Finance_Users -Jess Gomez



Sales_Users- Mike Hoyes

6.DNS and Domain Verification

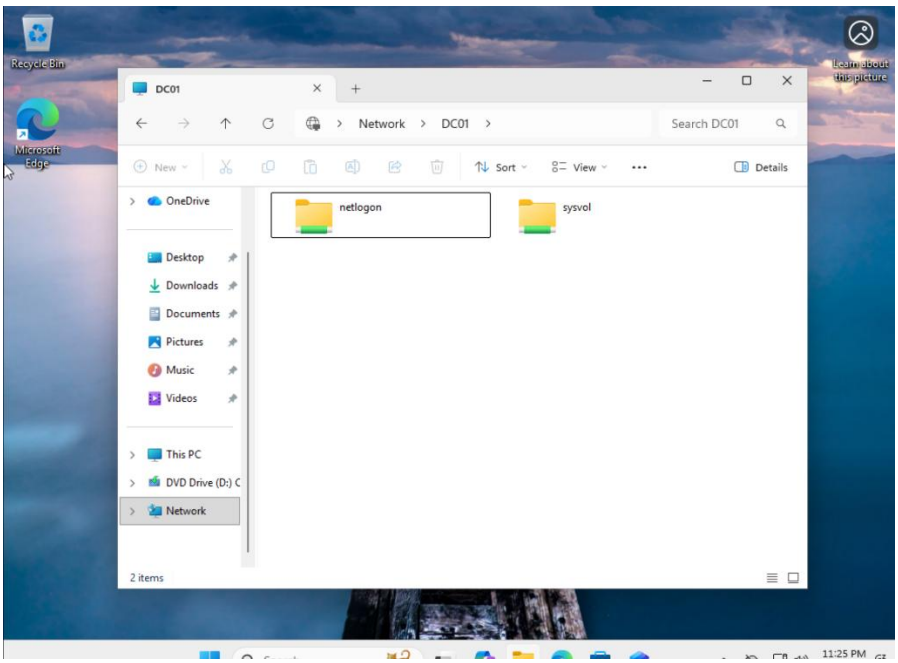
I set up DNS on the domain controller and used the Windows 11 computer to make sure homelab.local could be found. I used nslookup to check that the domain resolved to the correct IP address.



nslookup confirms that homelab.local resolves to 192.168.205.131

7.Domain Infrastructure Artifacts

I checked the domain controller for the SYSVOL and NETLOGON shares. These are created as part of the domain controller setup and give me another way to verify that the domain is working correctly.



DC01 showing the NETLOGON and SYSVOL shares.

Technical Skills

Windows Server 2022	Server setup and administration
Active Directory	Domain controller setup and directory management
DNS	Name resolution and basic domain troubleshooting
Users & Groups	User creation, password changes, and group membership
Organizational Units	Organizing users by department
Windows 11	Clientside DNS and domain testing
VMware	Running the virtual lab environment